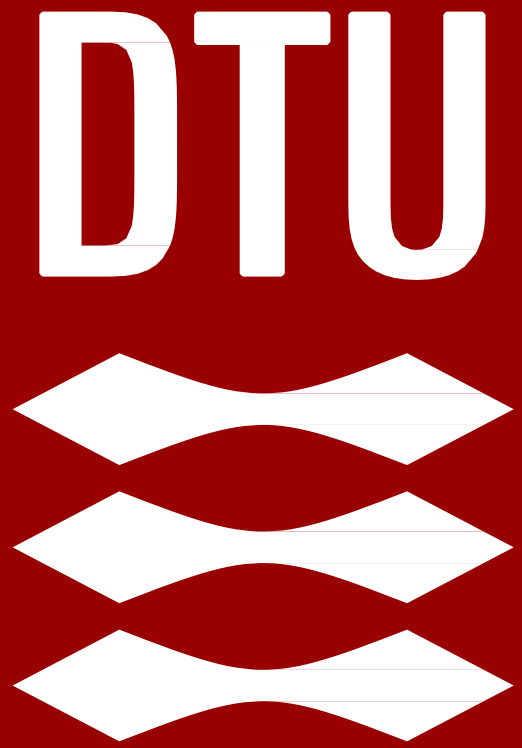

DTU

02526 - Mathematical Modelling - Prof. Dimitrios Papadopoulos

Exercise 3
Qattara Depression

Status

- **Today 17 March:** Start working on Exercise 3: Qattara Depression

- **Sunday 5 April:** Hand-in Exercise 3: Qattara Depression

Qattara Depression



Filling the Qattara Depression with water



Build a sea in the middle of Sahara Dessert

Initial idea: Dr. John Ball in 1927

Friedrich Bassler solution:
Build a canal from the Mediterranean Sea, 1964

Filling the Qattara Depression with water



Figure 2: Sampled height locations. Google maps.

- Estimate how much dirt needs to be removed
- Canal points in channel_data.txt
- Remove dirt by nuclear detonations

Minimize bombs

- *Problem with fallout*
- *Nuclear bombs are expensive*
- *Formulate is as a Mixed Integer Linear Problem*

Minimize bombs

$$\min \sum_{i=1}^{n} x_i \tag{1}$$

x_i: binary variable whether a nuke is planted in position i

Minimize bombs

$$\min \sum_{i=1}^{n} x_i \tag{1}$$

subject to

$$R_i \geq H_i + CHD \quad i = 1, \ldots, n \tag{2}$$

x_i: binary variable whether a nuke is planted in position i

R_i: amount of dirt removed in position i

H_i: height in position i

CHD: necessary channel depth

Minimize bombs

$$\min \sum_{i=1}^{n} x_i \tag{1}$$

subject to

$$R_i \geq H_i + CHD \quad i = 1, \ldots, n \tag{2}$$

$$R_i = K_2 \cdot x_{i-2} + K_1 \cdot x_{i-1} + K_0 \cdot x_i + K_1 \cdot x_{i+1} + K_2 \cdot X_{i+2} \quad i = 1, \ldots, n \tag{3}$$

$$x_i \in \{0, 1\} \quad i = 1, \ldots, n \tag{4}$$

x_i: binary variable whether a nuke is planted in position i

R_i: amount of dirt removed in position i

H_i: height in position i

CHD: necessary channel depth

K_0: amount of dirt removed in specific location

K_1: amount of dirt removed in neighboring positions

K_2: amount of dirt removed 2 positions away

Minimize bombs

$$\min \sum_{i=1}^{n} x_i \tag{1}$$

subject to

$$R_i \geq H_i + CHD \quad i = 1, \ldots, n \tag{2}$$

$$R_i = K_2 \cdot x_{i-2} + K_1 \cdot x_{i-1} + K_0 \cdot x_i + K_1 \cdot x_{i+1} + K_2 \cdot X_{i+2} \quad i = 1, \ldots, n \tag{3}$$

$$x_i \in \{0, 1\} \quad i = 1, \ldots, n \tag{4}$$

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K_2: amount of dirt removed 2 positions away

Different Solutions – Problems 3,4,5,6

- Direct solution (JULIA nukeDemo.jl with Cbc and GLPK solver)
- Smooth solution
- Placement constraints
- Dial a yield

Next steps

- **Build a sea in Sahara:** https://www.youtube.com/watch?v=8t3X7tUTk5o
- **Project Sedan – Nuclear bombs:** https://www.youtube.com/watch?v=mB9-1XMVpsY
- **Install JULIA**
- **Start working and Good luck :)**

Questions??